

Response to Assessment (github link and conceptual answers).

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SOLUTION TO CODING QUESTIONS

The solutions to the coding questions can be found in my github repository: https://github.com/FisayoD/blockhouse_assessment.git

Dear Hiring Manager,

Thank you for the opportunity. Please find below my responses to the assessment questions.

1. What's the motivation behind measuring OFI at multiple depth levels of the order book?

The motivation behind measuring OFI at multiple depth levels of the order book is that deeper levels contain valuable information about price movements that isn't captured by just looking at the best level. The research shows that for certain stocks (particularly those with high trading volume and low volatility), deeper levels in the order book influence price movements more than the best level.

Researchers can create a more complete picture of market supply and demand by examining multiple levels. While the best level shows immediate executable orders, deeper levels reveal potential liquidity that could become available as prices move, offering insight into how market participants are positioned at various price points.

The integrated OFI measure that combines information from multiple levels provides superior explanatory power for price movements compared to using only best-level OFI. This is particularly important when analyzing the relationship between order flow in one stock and price movements in other stocks (cross-impact analysis).

2. Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

The authors use Lasso regression rather than OLS for estimating cross-impact because OLS becomes statistically unsuitable when analyzing numerous assets simultaneously. With approximately 100 stocks but only 30 data points in typical estimation windows, OLS fails as the parameters outnumber observations. The significant correlation between order flows across stocks creates multicollinearity issues that violate key OLS assumptions.

Market dynamics naturally exhibit sparsity, with only a few stocks meaningfully impacting any particular stock. Lasso addresses this by automatically identifying significant relationships while eliminating negligible influences through coefficient shrinkage. This produces cleaner, more interpretable results that better represent the actual market structure.

The resulting sparse coefficient matrices enable more effective visualization and analysis of cross-asset relationships, revealing important sectoral patterns and specific company interactions. These statistical and practical advantages make Lasso the appropriate methodological choice for robust cross-impact estimation in the complex, high-dimensional setting of financial markets.

3. Why is OFI considered a better predictor of short-term returns than trade volume?

OFI is considered a better predictor of short-term returns than trade volume because it captures the directional pressure of order flow, measuring the imbalance between buying and selling intentions. According to the paper, Cont et al. (2014) found “that over short time intervals, price changes are mainly driven by the Order Flow Imbalance” and Kolm et al. (2023) demonstrated that “forecasting deep learning models trained on OFIs significantly outperform most models trained directly on order books or returns.”

Unlike trade volume, which only reflects completed transactions without indicating direction, OFI captures changes in the limit order book, including additions, cancellations, and executions. This provides a more comprehensive view of supply and demand dynamics. OFI explicitly measures buying versus selling pressure, while volume measures activity level regardless of direction.

The paper’s results confirm OFI’s predictive advantage by showing that OFI-based models generate profitable trading strategies, particularly over short time horizons of several minutes. The authors demonstrate that lagged cross-asset OFIs improve forecasting of future returns, though this predictability decays rapidly over longer time horizons.